

SANSKAR SHIMPI

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EDUCATION

F-1 Student · OPT Eligible May 2028

New York University, Tandon School of Engineering

M.S. Computer Science · Distributed Systems, ML Systems, Advanced ML, Big Data

Brooklyn, NY

Sept 2026 – May 2028

Vellore Institute of Technology

B.Tech Computer Science and Engineering — GPA: 3.68/4.0

Vellore, India

Sept 2022 – April 2026

EXPERIENCE

Research Engineer, Computer Vision

Jan 2025 – Jun 2026

ISRO–VIT Collaborative Research Program

Vellore, India

- Production-ready CV system for ISRO aerospace weld radiography: multi-stage image enhancement (CLAHE, sharpening, noise reduction) feeding YOLOv8 trained 300+ epochs at 640×640 ; mAP@0.5 = 0.883, precision 0.890, F1 0.86 across porosity, LOF, and linear pore defect classes.
- 70/15/15 stratified augmentation; per-defect ROI feature extraction (entropy, skewness, kurtosis), MySQL-backed traceability; ISRO LPSC Letter of Appreciation (B. Ramprasad, Manager QC) for core system development.

Backend Developer Intern

Jun 2024 – Jul 2024

Acumen Advertising DMCC

Dubai, UAE

- JWT auth layer for VoucherSkout's payment integrations (Samsung Pay, Noon Pay, Emirates NBD); reduced p50 latency 30% (200→150 ms) and error rate 40% across 2K+ daily transactions; raised unit-test coverage to 70%.

PROJECTS

DriftSentinel | Python · Kafka · Spark · scikit-learn · MLflow · Docker · Kubernetes · Terraform · AWS

Aug 2025 – Nov 2025

- End-to-end ML observability platform on Kafka + Spark Structured Streaming: 5-method parallel drift consensus (KS, χ^2 , PSI, Jensen-Shannon, SHAP-Delta); 0.45-hr MTTD, $9 \times$ faster than 4-hr baseline; cost-aware retraining selector (auditable decision tree + business-rule overrides) routes each incident to one of four strategies under a compute-cost ceiling.
- SPRT-gated canary promotion (5%→20%→50%→100%) with auto-rollback on latency/error regression; 121-test suite, 90–100% coverage on core modules; Kubernetes manifests, Terraform IaC, MLflow tracking, Grafana/Prometheus, GitHub Actions CI (Python 3.11/3.12 matrix).

AgentMemOS | Python · asyncio · gRPC · FastAPI · Redis · Pinecone · Neo4j · PostgreSQL · Kafka · OPA

Apr 2026 – Jun 2026

- 4-tier LLM agent memory OS (Redis/Pinecone/Neo4j/PostgreSQL): parallel cross-tier read fan-out, 5-signal importance-scored write router; async gRPC data plane at 5,537 req/s, P95 2.23 ms; semantic-LRU eviction with ghost tombstones for cold-tier prefetch; Kafka WAL for durable writes.
- OPA/Rego federation engine enforcing cross-agent access control and PII field redaction; 100-test suite (property-based, concurrency, REST integration, proto-contract) via in-memory fakes; GitHub Actions CI with ruff lint and coverage gate.

LineageIQ | Python · FastAPI · Neo4j · Kafka · PostgreSQL · Redis · React

Jan 2026 – Mar 2026

- Column-level lineage engine parsing dbt, Spark, SQL, and Kafka metadata into Neo4j; blast-radius analyzer scores schema-change impact by SLA tier, freshness cadence, and downstream fan-out; 1,389 req/s, 44 ms P95 under 50-VU k6 load.
- Text2Cypher NL layer with static schema validator blocking hallucinated labels and write operations before execution; 93 passing tests (Hypothesis property-based, Cypher-injection/security suites); 3-stage GitHub Actions CI.

OPEN SOURCE

LangChain-Neo4j PR#119 (Merged) · **MLflow** PR#23147 (+103 lines) · **Apache Airflow** PR#66648 (+64 lines) · **FastAPI** PR#15496 — bug fix, features, regression suites; all CI checks passed.

COMPETITIONS

IMC Prosperity 4 (2026) — Top **4%** Manual Trading, top **6%** overall among 18,800+ global teams; team *Wolf of All Streets*.

SKILLS

Languages: Python, C++, SQL

ML & MLOps: PyTorch, TensorFlow, scikit-learn, MLflow, SHAP, YOLOv8, Pinecone, LangChain, LLMs, RAG

Systems & Backend: Kafka, Spark, FastAPI, gRPC, asyncio, Redis, Neo4j, PostgreSQL, OPA/Rego

DevOps & Cloud: Docker, Kubernetes, Terraform, Airflow, Prometheus, Grafana, OpenTelemetry, GitHub Actions, AWS

Foundations: Distributed Systems, System Design, Algorithms & Data Structures, Linux, Git